

Two-way fixed effects estimators with heterogeneous treatment effects*

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Abstract

Linear regressions with period and group fixed effects are widely used to estimate treatment effects. We show that they estimate weighted sums of the average treatment effects (ATE) in each group and period, with weights that may be negative. Due to the negative weights, the linear regression coefficient may for instance be negative while all the ATEs are positive. We propose another estimator that solves this issue. In the two applications we revisit, it is significantly different from the linear regression estimator.

Keywords: linear regressions, fixed effects, heterogeneous treatment effects, difference-in-differences.

JEL Codes: C21, C23

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Our paper is related to the DID literature. Our main result generalizes Theorem 1 in de Chaisemartin and D’Haultfoeuille (2018). When the data has two groups and two periods, the Wald-DID estimand considered therein is equal to β_{fe} and β_{fd} . Our results on β_{fe} and β_{fd} are thus extensions of that theorem to the case with multiple periods and groups.³ Moreover, our DID_M estimator is related to the Wald-TC estimator with many groups and periods proposed in de Chaisemartin and D’Haultfoeuille (2018), and to the multi-period DID estimator proposed by Imai and Kim (2018). In Section 4, we explain the differences between those three estimators.

More recently, Borusyak and Jaravel (2017), Abraham and Sun (2018), Athey and Imbens (2018), Callaway and Sant’Anna (2018), and Goodman-Bacon (2018) study the special case of staggered adoption designs, where the treatment of a group is weakly increasing over time. Those papers derive some important results specific to that design that we do not consider here. Still, some of the results in those papers are related to ours, and we describe precisely those connections later in the paper. The most important dimension on which our paper differs from those is that our results apply to any two-way fixed effects regressions, not only to those with staggered adoption. In our survey of the AER papers estimating two-way fixed effects regressions, less than 10% have a staggered adoption design. This suggests that while staggered adoptions are an important research design, they may account for a relatively small minority of the applications where two-way fixed effects regressions have been used.

The paper is organized as follows. Section 2 introduces the set-up. Section 3 presents our decomposition results. Section 4 introduces our alternative estimator. Section 5 briefly describes some of the extensions covered in our Web Appendix. Section 6 presents our survey of the articles published in the AER, and our two empirical applications.

2 Set up

One considers observations that can be divided into G groups and T periods. For every $(g, t) \in \{1, \dots, G\} \times \{1, \dots, T\}$, let $N_{g,t}$ denote the number of observations in group g at period t , and let $N = \sum_{g,t} N_{g,t}$ be the total number of observations. The data may be an individual-level panel or repeated cross-section data set where groups are, say, individuals’ county of birth. The data could also be a cross-section where cohort of birth plays the role of time. For instance, Duflo (2001) compares the schooling of different cohorts in Indonesia, some of which were exposed to a school construction program. It is also possible that for all (g, t) , $N_{g,t} = 1$, e.g. a group is one individual or firm. All of the above are special cases of the data structure we consider.

One is interested in measuring the effect of a treatment on some outcome. Throughout the paper

³In fact, a preliminary version of our main result appeared in a working paper version of de Chaisemartin and D’Haultfoeuille (2018) (see Theorems S1 and S2 in de Chaisemartin and D’Haultfoeuille, 2015).

denote the number of observations with treatment d' at period $t - 1$ and d at period t . Let

$$\begin{aligned} \text{DID}_{+,t} &= \sum_{g:D_{g,t}=1,D_{g,t-1}=0} \frac{N_{g,t}}{N_{1,0,t}} (Y_{g,t} - Y_{g,t-1}) - \sum_{g:D_{g,t}=D_{g,t-1}=0} \frac{N_{g,t}}{N_{0,0,t}} (Y_{g,t} - Y_{g,t-1}), \\ \text{DID}_{-,t} &= \sum_{g:D_{g,t}=D_{g,t-1}=1} \frac{N_{g,t}}{N_{1,1,t}} (Y_{g,t} - Y_{g,t-1}) - \sum_{g:D_{g,t}=0,D_{g,t-1}=1} \frac{N_{g,t}}{N_{0,1,t}} (Y_{g,t} - Y_{g,t-1}). \end{aligned}$$

Note that $\text{DID}_{+,t}$ is not defined when there is no group such that $D_{g,t} = 1, D_{g,t-1} = 0$, or no group such that $D_{g,t} = 0, D_{g,t-1} = 0$. In such instances, we let $\text{DID}_{+,t} = 0$. Similarly, let $\text{DID}_{-,t} = 0$ when there is no group such that $D_{g,t} = 1, D_{g,t-1} = 1$ or no group such that $D_{g,t} = 0, D_{g,t-1} = 1$. Finally, let

$$\text{DID}_M = \sum_{t=2}^T \left(\frac{N_{1,0,t}}{N_S} \text{DID}_{+,t} + \frac{N_{0,1,t}}{N_S} \text{DID}_{-,t} \right).$$

Theorem 3 *If Assumptions 1, 2, 4, 5, and 9-12 hold, $E[\text{DID}_M] = \delta^S$.*

In Section ?? of the Web Appendix, we also show that when G goes to infinity, DID_M is a consistent and asymptotically normal estimator of δ^S . The DID_M estimator is computed by the `fuzzydid` and `did_multipligt` Stata packages.

Here is the intuition underlying Theorem 3. $\text{DID}_{+,t}$ compares the evolution of the mean outcome between $t - 1$ and t in two sets of groups: the joiners, and those remaining untreated. Under Assumptions 4 and 5, $\text{DID}_{+,t}$ estimates the joiners' treatment effect. Similarly, $\text{DID}_{-,t}$ compares the evolution of the outcome between $t - 1$ and t in two sets of groups: those remaining treated, and the leavers. Under Assumptions 9 and 10, it estimates the leavers' treatment effect. Finally, DID_M is a weighted average of those DID. Note that in staggered designs, there are no groups whose treatment decreases over time, so DID_M is only a weighted average of the $\text{DID}_{+,t}$ estimators. Note also that one can separately estimate the joiners' and the leavers' treatment effect, by computing separately weighted averages of the $\text{DID}_{+,t}$ and $\text{DID}_{-,t}$ estimators. The former estimator only relies on Assumptions 4 and 5, while the latter only relies on Assumptions 9 and 10.

DID_M is related to two other estimators. First, it is related to the Wald-TC estimator in point 2 of Theorem S1 in the Web Appendix of de Chaisemartin and D'Haultfœuille (2018), but the weighting of $\text{DID}_{+,t}$ and $\text{DID}_{-,t}$ therein differs. As a result, DID_M estimates Δ^S under weaker assumptions. DID_M is also related to the multi-period DID estimator in Imai and Kim (2018). However, the multi-period DID estimator is a weighted average of the $\text{DID}_{+,t}$, so it does not estimate the leavers' treatment effect, and applies to a smaller population. Besides, Imai and Kim (2018) do not establish the properties of their estimator. Finally, they do not generalize it to non-binary treatments, something we do in Section ?? of the Web Appendix.